



Sustainable Development SDG 13

# Global Climate Events and Economic Impact Dataset

From Q1 2020 To Q3 2025



GLOBAL CLIMATE EVENTS



ECONOMIC IMPACT  
DATA ANALYSIS & INSIGHTS



## Prepared by Group (1)



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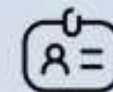
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# Agenda



1- Introduction

2- Objectives

3- Data Overview

4- Methodology

5- Insights & Recommendations



# 1- Introduction



This project utilizes a **secondary dataset** sourced from Kaggle, acknowledging the typical limitations associated with non-primary data sources. The analysis is based on approximately 3,000 **observational records**, aiming to explore underlying **patterns and relationships** within the data.

The project focuses on the multidimensional nature of natural disasters by considering **20 variables** covering **human, economic, and infrastructure**-related dimensions, with the objective of providing an initial analytical understanding of the complexity of **disaster events**.



# 2- 1 Objectives



This project aims to analyze the relationships between key variables associated with natural disasters, with a particular focus on the interactions between human impact (affected population), economic consequences, and infrastructure damage.

It also seeks to move beyond a univariate perspective by providing an integrated analytical framework that supports decision-makers in developing a more comprehensive understanding of disaster events and improving priority-setting in crisis management.

Furthermore, the project emphasizes data-driven decision-making by reducing reliance on purely economic indicators and incorporating the human dimension into impact assessment and resource allocation.



# 2- 2 Objectives



The project is aligned with the Sustainable Development Goals (SDGs), particularly:

- SDG 13 – Climate Action: improving understanding and response to climate-related disasters.
- SDG 11 – Sustainable Cities and Communities: enhancing resilience to disaster risks.
- SDG 10 – Reduced Inequalities: promoting fairness in humanitarian aid distribution.
- SDG 16 – Peace, Justice and Strong Institutions: supporting transparency and crisis governance.

Sustainable Development Goals (SDGs)





# 3- Data Overview



- Source: Kaggle – Global Climate Events and Economic Impact Dataset
- Observations: 3,000 records
- Variables: 20 features
- Scope: 12 types of natural disaster events
- Dimensions: Human, economic, infrastructure impact
- Purpose: Exploratory data analysis & pattern identification to support decision-making.



# 4- Methodology

## Data Cleaning & Feature Engineering

### ► Severity Score Transformation

(1-10 Scale) → Low Severity | Medium Severity | High Severity

### ► Response Time Categorization

Hours → Very Fast Response | Fast Response | Moderate Response | Slow Response

### ► Infrastructure Damage Classification

(0-100 Scale) → Low Damage | Moderate Damage | High Damage | Critical Damage

### ► Seasonal Feature Engineering

Created a new categorical feature representing Seasons:

Winter | Spring | Summer | Autumn

### ► Geographic Data Correction

Corrected invalid Latitude & Longitude values using official country coordinate references from Google Developers Public Data.



### ► Geographic Mapping

Mapped corrected coordinates with the original dataset using Country Names as the common matching field.



| Event_id | Date      | Year | Month Name | Season | Country        | Event_Type        | Severity_Level  | Duration_Day | Affected_Population | Economic_Impact_Million_US | Infrastructure_Damage_Type |
|----------|-----------|------|------------|--------|----------------|-------------------|-----------------|--------------|---------------------|----------------------------|----------------------------|
| EV01539  | 1/1/2020  | 2020 | January    | Winter | Japan          | Tsunami           | Low Severity    | 1            | 420956              | 0.01                       | Light Damage               |
| EV00608  | 1/22/2020 | 2020 | January    | Winter | Japan          | Flood             | High Severity   | 9            | 534076              | 4.8                        | Light Damage               |
| EV01783  | 1/22/2020 | 2020 | January    | Winter | Japan          | Hurricane         | Low Severity    | 1            | 208674              | 0.04                       | Light Damage               |
| EV02303  | 1/1/2020  | 2020 | January    | Winter | Qatar          | Hurricane         | Low Severity    | 4            | 3276                | 0                          | Light Damage               |
| EV00190  | 1/23/2020 | 2020 | January    | Winter | Qatar          | Drought           | Medium Severity | 43           | 24889               | 0.3                        | Light Damage               |
| EV01796  | 1/2/2020  | 2020 | January    | Winter | Canada         | Drought           | Low Severity    | 6            | 120382              | 0.1                        | Light Damage               |
| EV00175  | 1/2/2020  | 2020 | January    | Winter | Poland         | Heatwave          | Medium Severity | 16           | 185527              | 1.27                       | Light Damage               |
| EV01115  | 1/3/2020  | 2020 | January    | Winter | UAE            | Wildfire          | Medium Severity | 16           | 176642              | 2.01                       | Light Damage               |
| EV00057  | 1/3/2020  | 2020 | January    | Winter | Colombia       | Cold Wave         | Low Severity    | 72           | 246108              | 0.63                       | Light Damage               |
| EV02417  | 1/4/2020  | 2020 | January    | Winter | India          | Earthquake        | Low Severity    | 2            | 5076307             | 0.77                       | Light Damage               |
| EV01755  | 1/5/2020  | 2020 | January    | Winter | Kazakhstan     | Earthquake        | Medium Severity | 1            | 22405               | 0.01                       | Light Damage               |
| EV01104  | 1/11/2020 | 2020 | January    | Winter | Kazakhstan     | Volcanic Eruption | Low Severity    | 2            | 60545               | 0                          | Light Damage               |
| EV00333  | 1/5/2020  | 2020 | January    | Winter | Czech Republic | Landslide         | Low Severity    | 1            | 8681                | 0                          | Light Damage               |
| EV00121  | 2/2/2020  | 2020 | February   | Winter | Czech Republic | Landslide         | Medium Severity | 2            | 12131               | 0.04                       | Light Damage               |
| EV01124  | 1/5/2020  | 2020 | January    | Winter | Mexico         | Hailstorm         | Medium Severity | 1            | 972775              | 0.23                       | Light Damage               |
| EV00511  | 2/1/2020  | 2020 | February   | Winter | Mexico         | Heatwave          | Low Severity    | 26           | 578167              | 1.58                       | Light Damage               |
| EV02537  | 1/6/2020  | 2020 | January    | Winter | Finland        | Drought           | Low Severity    | 7            | 12691               | 0.01                       | Light Damage               |
| EV01457  | 1/17/2020 | 2020 | January    | Winter | Finland        | Volcanic Eruption | Low Severity    | 2            | 3561                | 0                          | Light Damage               |
| EV01632  | 1/31/2020 | 2020 | January    | Winter | Finland        | Drought           | Medium Severity | 26           | 62861               | 1.94                       | Light Damage               |
| EV01297  | 1/8/2020  | 2020 | January    | Winter | New Zealand    | Tsunami           | Medium Severity | 1            | 59222               | 0.02                       | Light Damage               |
| EV00096  | 1/9/2020  | 2020 | January    | Winter | Austria        | Heatwave          | Medium Severity | 51           | 56727               | 2.04                       | Light Damage               |
| EV02633  | 1/11/2020 | 2020 | January    | Winter | Italy          | Heatwave          | Medium Severity | 37           | 541332              | 26.98                      | Moderate Damage            |
| EV00444  | 1/12/2020 | 2020 | January    | Winter | Turkey         | Hurricane         | Low Severity    | 5            | 84244               | 0.02                       | Light Damage               |
| EV00848  | 1/13/2020 | 2020 | January    | Winter | Peru           | Wildfire          | Medium Severity | 18           | 137975              | 0.22                       | Light Damage               |
| EV02594  | 1/18/2020 | 2020 | January    | Winter | Peru           | Landslide         | Low Severity    | 2            | 156019              | 0                          | Light Damage               |
| EV00051  | 1/14/2020 | 2020 | January    | Winter | Iraq           | Volcanic Eruption | Low Severity    | 1            | 96854               | 0                          | Light Damage               |
| EV02691  | 1/18/2020 | 2020 | January    | Winter | Iraq           | Drought           | Medium Severity | 19           | 557975              | 0.27                       | Moderate Damage            |
| EV02170  | 1/15/2020 | 2020 | January    | Winter | Chile          | Heatwave          | Low Severity    | 5            | 79043               | 0.07                       | Light Damage               |
| EV02310  | 2/3/2020  | 2020 | February   | Winter | Chile          | Landslide         | Low Severity    | 1            | 141006              | 0.03                       | Light Damage               |
| EV00131  | 1/15/2020 | 2020 | January    | Winter | Belgium        | Hurricane         | Low Severity    | 1            | 55508               | 0.01                       | Light Damage               |
| EV02928  | 1/24/2020 | 2020 | January    | Winter | Belgium        | Flood             | Medium Severity | 3            | 58770               | 0.13                       | Light Damage               |
| EV02420  | 1/16/2020 | 2020 | January    | Winter | Philippines    | Volcanic Eruption | Low Severity    | 1            | 666137              | 0.01                       | Light Damage               |
| EV01721  | 1/28/2020 | 2020 | January    | Winter | Philippines    | Hurricane         | Medium Severity | 17           | 638994              | 0.04                       | Light Damage               |
| EV01411  | 1/31/2020 | 2020 | January    | Winter | Philippines    | Earthquake        | Medium Severity | 1            | 115733              | 0.01                       | Light Damage               |



# 4- Methodology

## Data Understanding - EDA

| <i>affected_population</i>                                     |             | <i>economic_impact_million_usd</i> |             | <i>international_aid_million_usd</i> |             |
|----------------------------------------------------------------|-------------|------------------------------------|-------------|--------------------------------------|-------------|
| Mean                                                           | 868550.4953 | Mean                               | 1.944366667 | Mean                                 | 0.05702     |
| Standard Error                                                 | 54949.16311 | Standard Error                     | 0.2872803   | Standard Error                       | 0.054879387 |
| Median                                                         | 178477      | Median                             | 0.09        | Median                               | 0           |
| Mode                                                           | 40289       | Mode                               | 0           | Mode                                 | 0           |
| Standard Deviation                                             | 3009689.615 | Standard Deviation                 | 15.73499007 | Standard Deviation                   | 3.005867817 |
| Sample Variance                                                | 9.05823E+12 | Sample Variance                    | 247.5899126 | Sample Variance                      | 9.035241333 |
| Kurtosis                                                       | 113.9899212 | Kurtosis                           | 1462.494732 | Kurtosis                             | 2990.50859  |
| Skewness                                                       | 9.373296544 | Skewness                           | 34.01289467 | Skewness                             | 54.64563059 |
| Range                                                          | 56247703    | Range                              | 718.21      | Range                                | 164.51      |
| Minimum                                                        | 622         | Minimum                            | 0           | Minimum                              | 0           |
| Maximum                                                        | 56248325    | Maximum                            | 718.21      | Maximum                              | 164.51      |
| Sum                                                            | 2605651486  | Sum                                | 5833.1      | Sum                                  | 171.06      |
| Count                                                          | 3000        | Count                              | 3000        | Count                                | 3000        |
| Mean -Median                                                   |             |                                    |             |                                      |             |
|                                                                |             | 690073.4953                        | 1.854366667 | 0.05702                              |             |
| Extreme Outliers-----> طبيعى ليس خطأ فى الإدخال لا يحتاج تعديل |             |                                    |             |                                      |             |
| Non-Parametric Inferential Statistics                          |             |                                    |             |                                      |             |

Detection of skewness & extreme outliers

# 4- Methodology

## Statistical Workflow

### Statistical Analysis

#### ► Descriptive Statistics

#### ► Inferential Statistics

- Spearman Correlation Analysis (Non-Parametric)
- Sensitivity Analysis
- Hypothesis Testing

### Statistical Notes

- Highly right-skewed data with extreme outliers
- Non-parametric statistical methods were selected
- Confidence Level = 95% ( $\alpha = 0.05$ )



## Why Spearman Correlation ?

# 4- Methodology



Correlation  $\neq$  Causation !



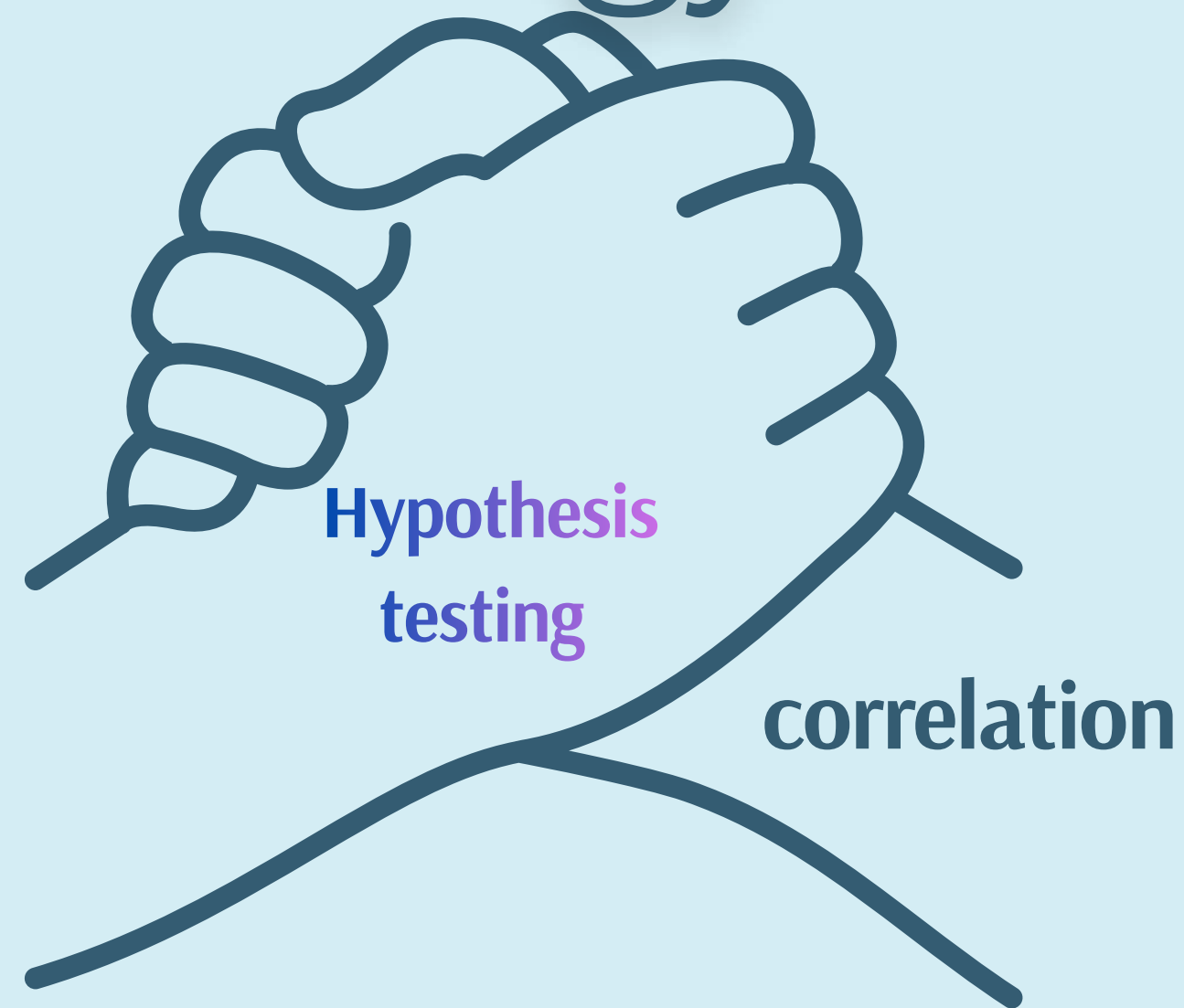
How to Prove  
**Causation** ??

Are these relationships **statistically valid** or just  
random correlation?





# 4- Methodology

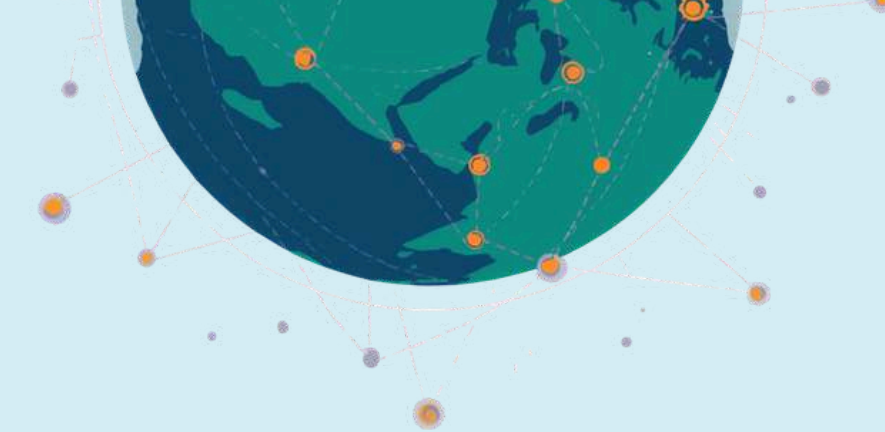


**Hypothesis testing goes beyond correlation to validate statistical significance.**



# 4- Methodology

## Statistical Workflow



1

Correlation Analysis

### Sensitivity Analysis

| Before Removing Outliers |                                   |           |                         | After Removing Outliers |                                   |         |                            |
|--------------------------|-----------------------------------|-----------|-------------------------|-------------------------|-----------------------------------|---------|----------------------------|
| Pearson Correlation      | Human Impact Vs Eco Impact        | 0.2206846 | طردية ضعيفة             | Pearson Correlation     | Human Impact Vs Eco Impact        | 0.24    | طردية ضعيفة                |
| Spearman Correlation     | Human Impact Vs Eco Impact        | 0.5431533 | طردية متوسطة            | Spearman Correlation    | Human Impact Vs Eco Impact        | 0.42    | طردية متوسطة               |
| Pearson Correlation      | Human Impact & International Aids | 0.050688  | ضعيفة جداً - شبه منعدمة | Pearson Correlation     | Human Impact & International Aids | #DIV/0! | لا يوجد رقم نفس بناءً عليه |
| Spearman Correlation     | Human Impact & International Aids | 0.0437965 | ضعيفة جداً - شبه منعدمة | Spearman Correlation    | Human Impact & International Aids | #DIV/0! | لا يوجد رقم نفس بناءً عليه |
| Pearson Correlation      | Eco Impact & International Aids   | 0.8368644 | طردية قوية جداً         | Pearson Correlation     | Eco Impact & International Aids   | #DIV/0! | لا يوجد رقم نفس بناءً عليه |
| Spearman Correlation     | Eco Impact & International Aids   | 0.0447817 | ضعيفة جداً - شبه منعدمة | Spearman Correlation    | Eco Impact & International Aids   | #DIV/0! | لا يوجد رقم نفس بناءً عليه |

The sensitivity analysis shows that correlations are highly affected by outliers. While a strong Pearson correlation was observed between Eco Impact and International Aids before cleaning, this relationship disappears when using Spearman correlation, indicating it is driven by extreme values rather than a stable monotonic relationship. After outlier removal, only a weak-to-moderate positive relationship remains between Human Impact and Eco Impact, suggesting a limited but more robust association between the variables.

## References

1- [Lumen Learning – Hypothesis Test for Correlation](#)

2- [Minitab – Methods and Formulas for Correlation](#)

### 2- 1

Hypothesis Testing for Correlation

#### 1- Spearman Hypothesis Test (Primary)

|             |              |
|-------------|--------------|
| t-statistic | 35.41999098  |
| P - Value   | 7.06959E-230 |

$H_0$ : لا توجد علاقة ( $\rho = 0$ )

$H_1$ : توجد علاقة ( $\rho \neq 0$ )

$$t = \frac{r_s \sqrt{n-2}}{\sqrt{1-r_s^2}}$$

t-test for Pearson Correlation

t-statistic

$$p = 2 \times P(T \geq |t|)$$

Spearman correlation =  $r_s$  •  
عدد البيانات =  $n$  •

### 2- 2

Hypothesis Testing for Correlation

#### 1- Spearman Hypothesis Test (Primary)

|             |              |
|-------------|--------------|
| t-statistic | 22.54513557  |
| P - Value   | 5.00146E-102 |

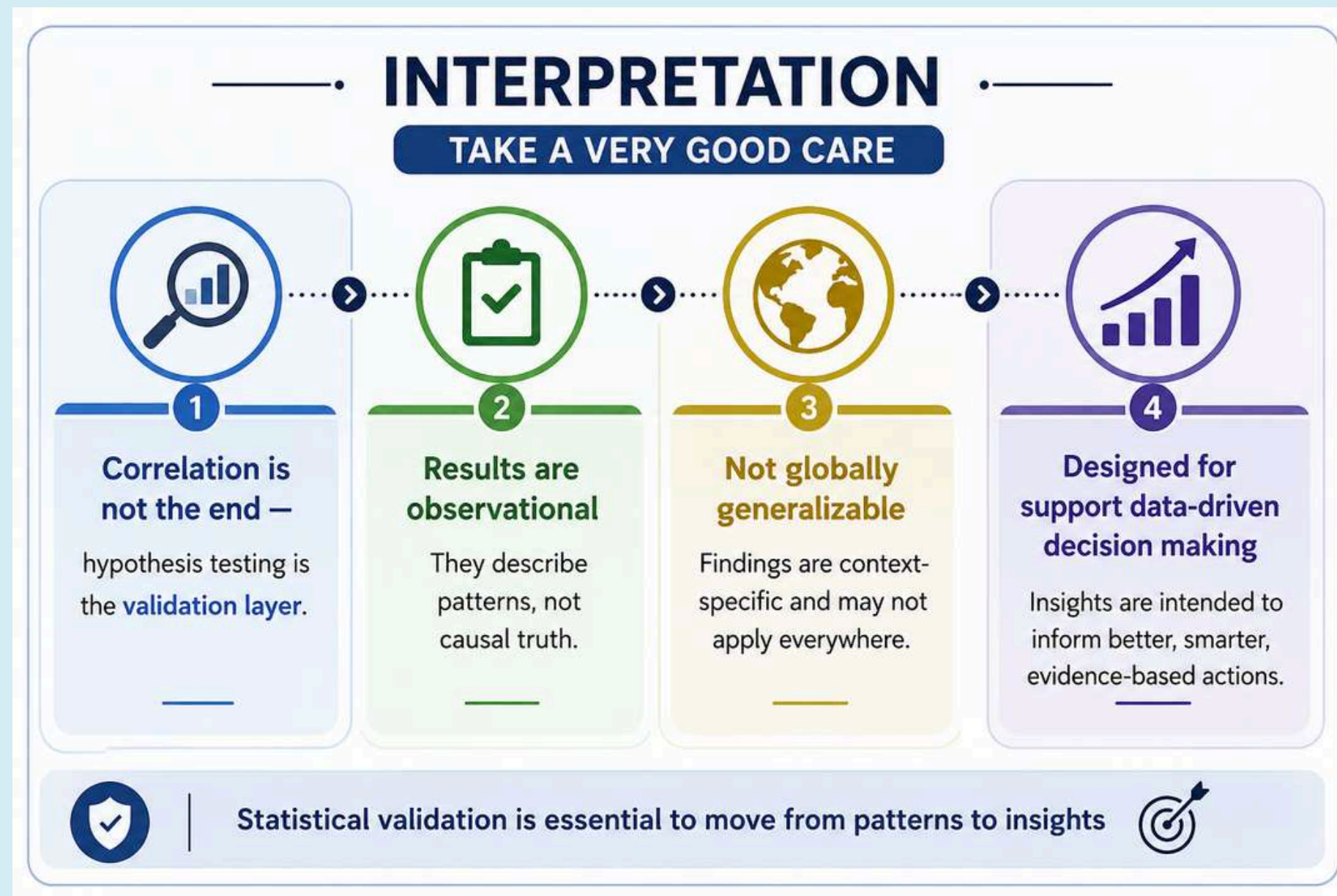
The Spearman correlation between Human Impact and Eco Impact remains positive and statistically highly significant both before and after outlier removal. However, the decrease in correlation strength after removing outliers suggests that extreme values were inflating the magnitude of the relationship.

Despite this, the persistence of a moderate correlation indicates a robust underlying monotonic association between the two variables.



# 4- Methodology

## Conclusion





# 4- Methodology

## DATA Modeling

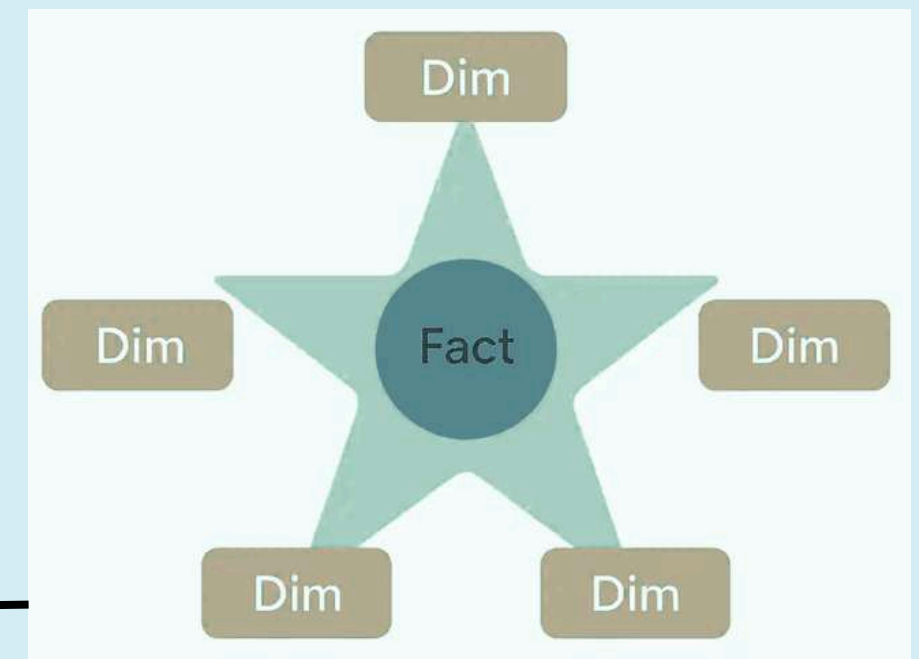
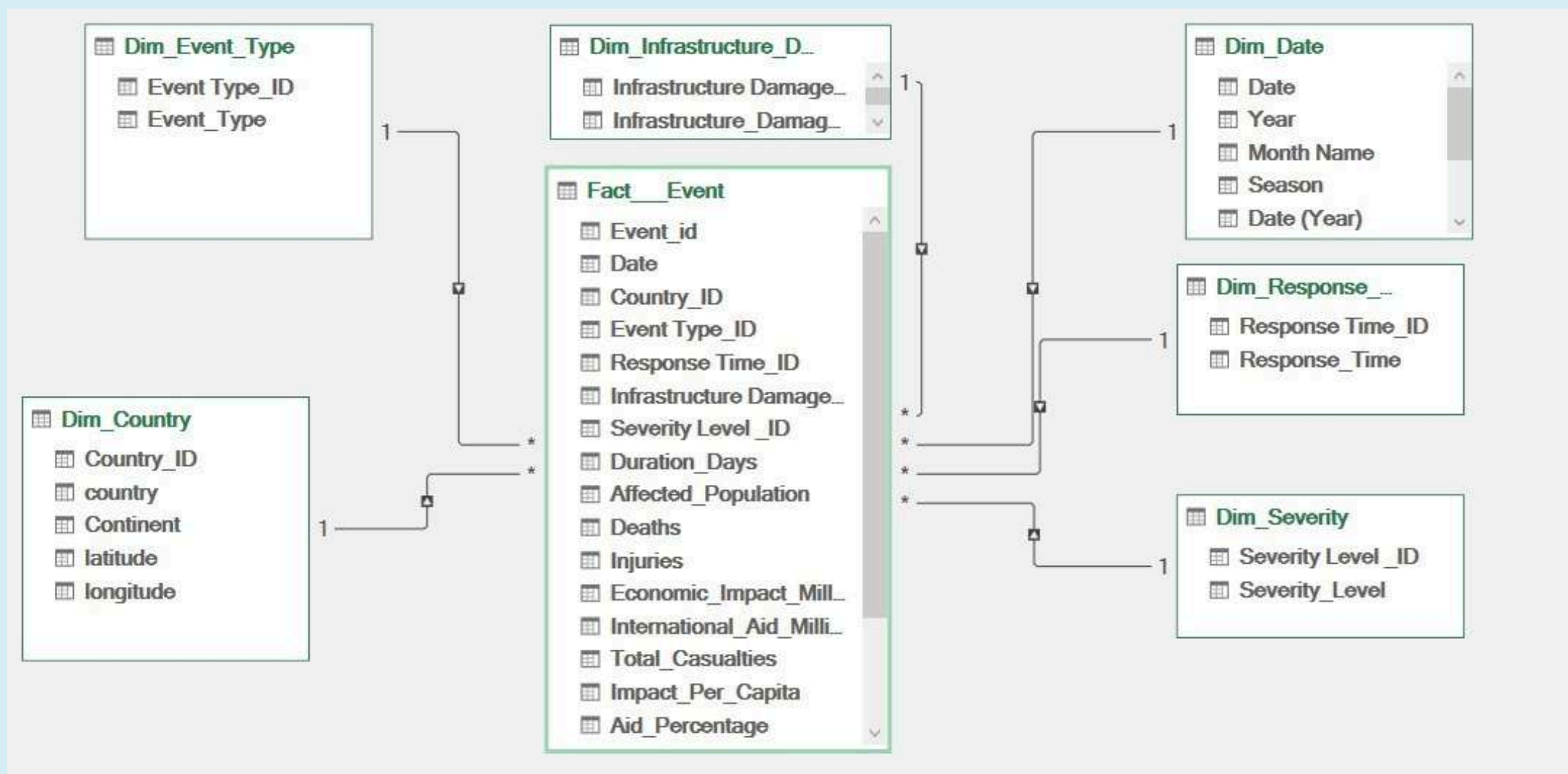


We moved from understanding the data to building analytical models that turn raw data into actionable insights.



# 4- Methodology

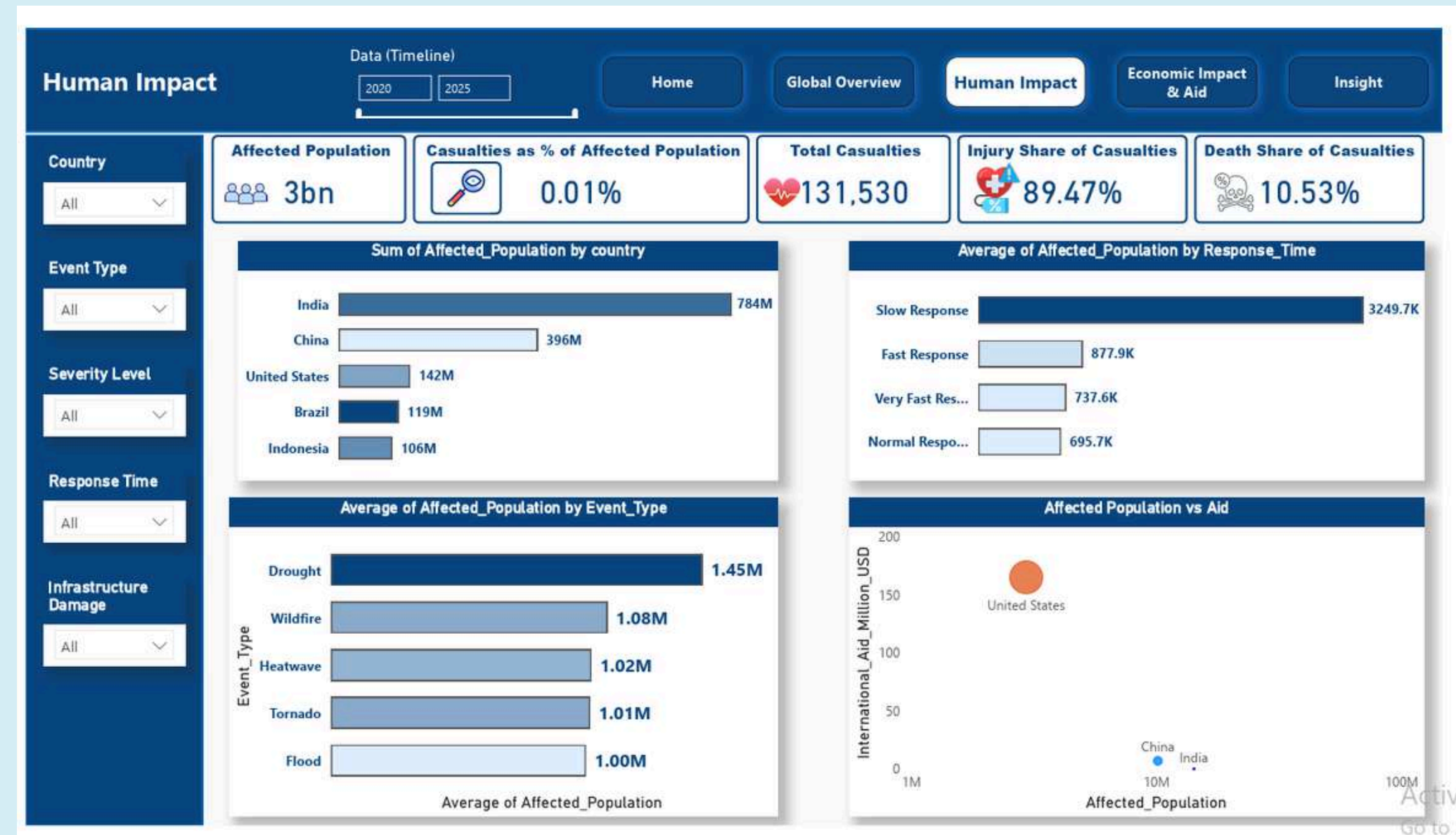
## DATA Modeling



Star Schema

# 4- Methodology

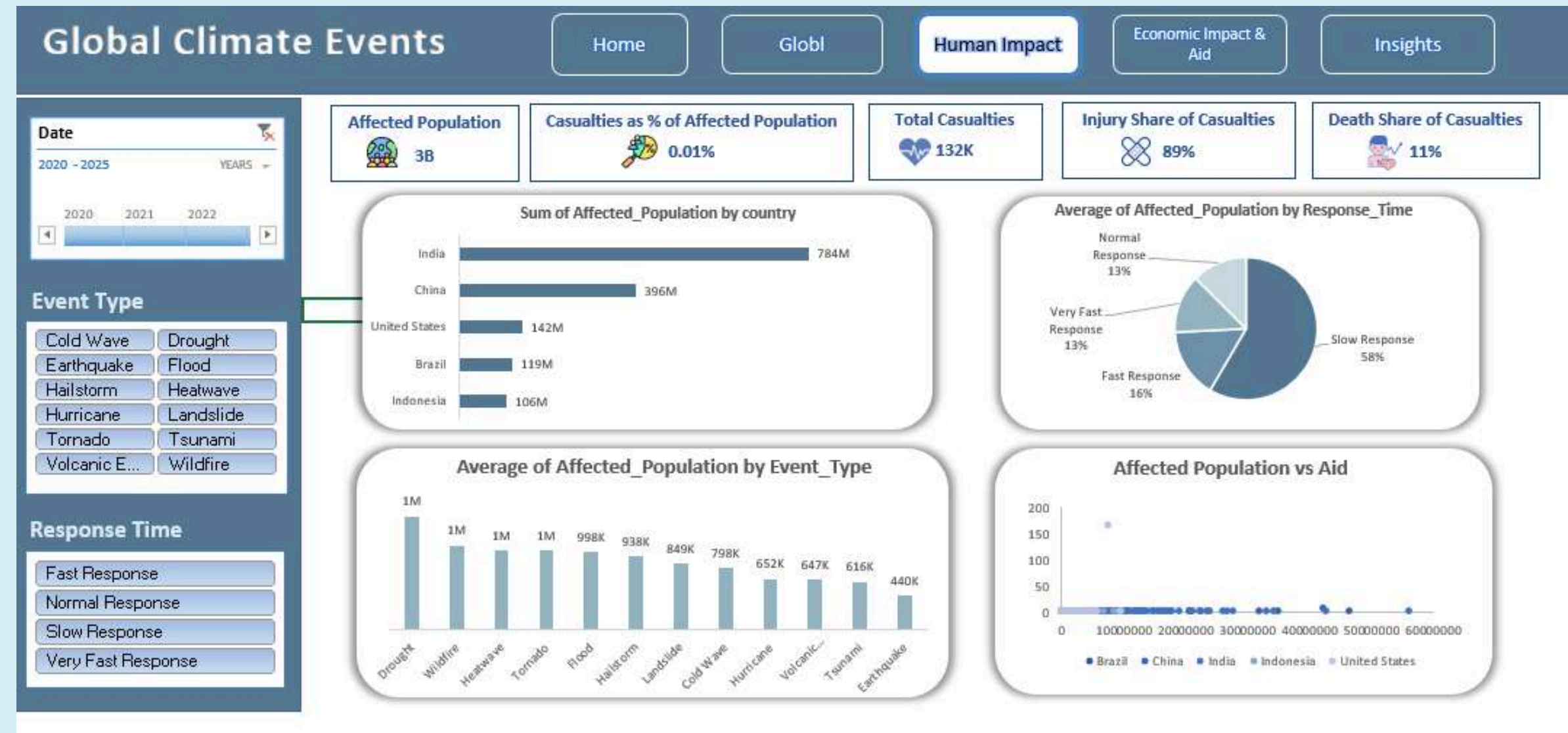
## Data Visualizations





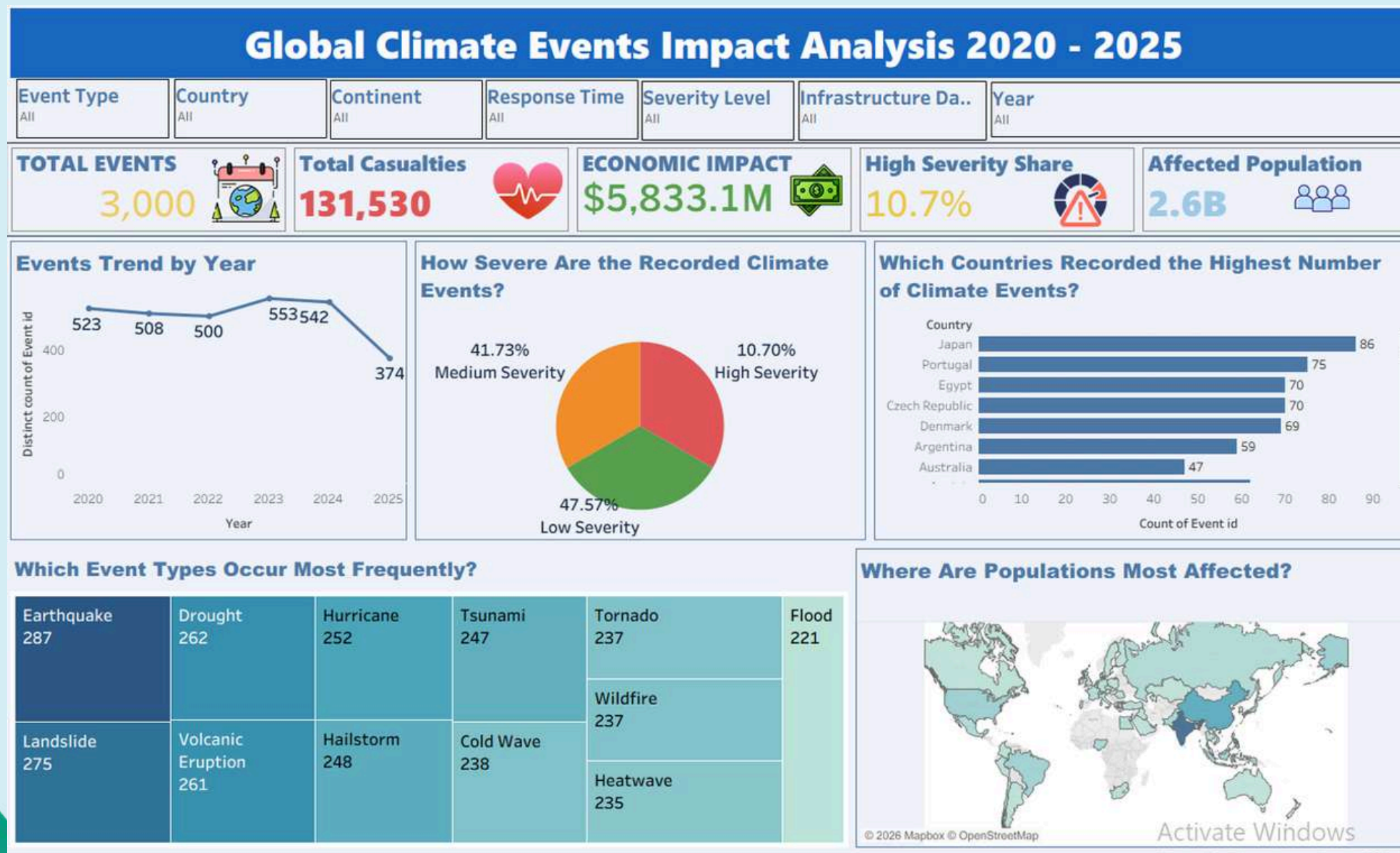
# 4- Methodology

## Data Visualizations



# 4- Methodology

## Data Visualizations



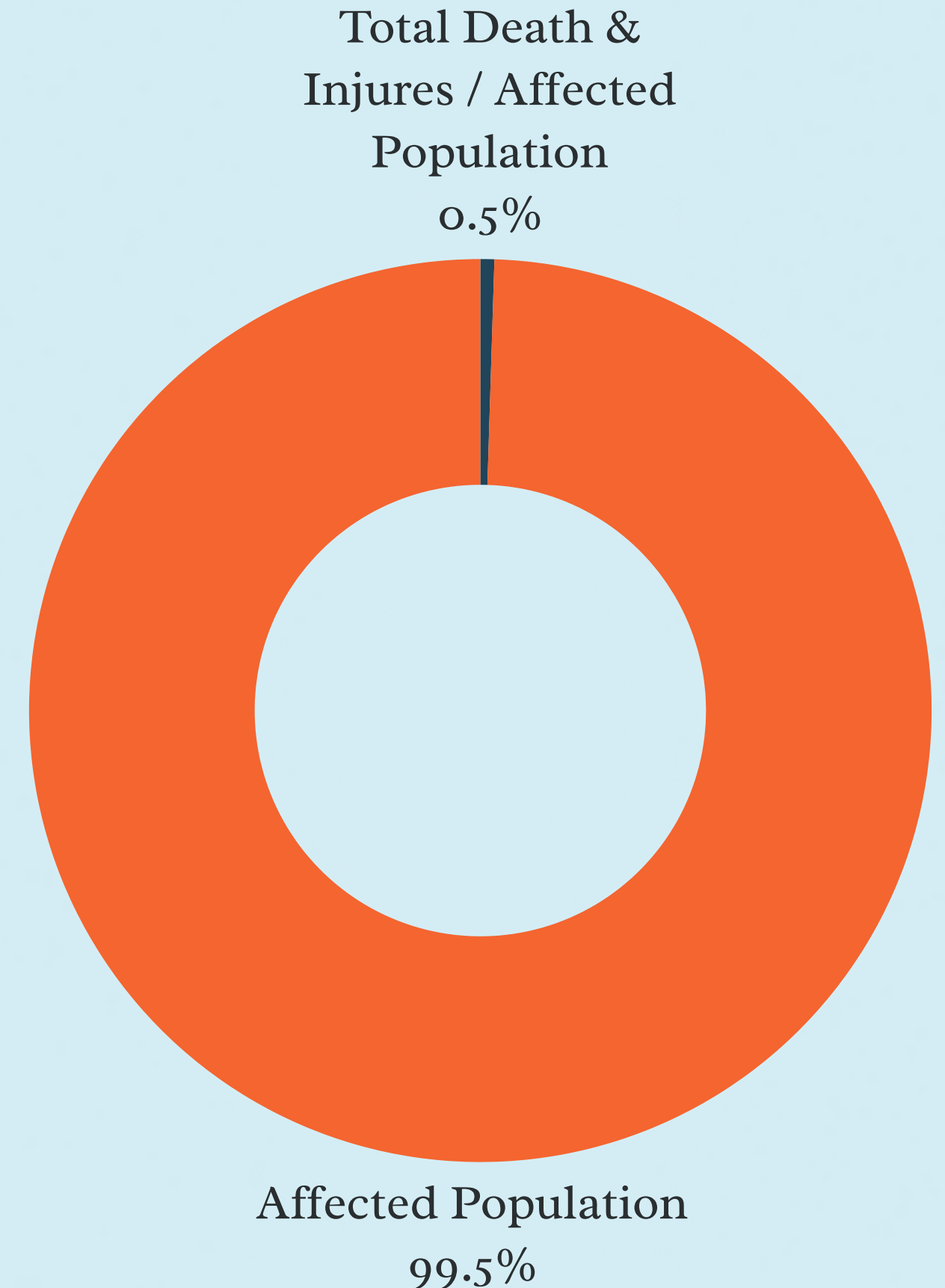


# Disaster Impact on People

Disaster impact is not only about deaths, but also about displacement and **human suffering**.



Insight





# 5- Insights & Recommendations

## KEY INSIGHTS

Evidence from Disaster Impact and Aid Analysis



01

### NATURE OF DISASTERS

Disasters are primarily socio-economic rather than lethal. Deaths and injuries account for less than 1% of the affected population.

The main impact is on livelihoods, displacement, and infrastructure disruption.



02

### HUMAN VS ECONOMIC IMPACT

There is a moderate relationship between affected population and economic losses.

Disasters that impact more people tend to generate higher economic damage, even when mortality remains relatively low.



03

### INFRASTRUCTURE SENSITIVITY

Disaster severity is one of the strongest drivers of infrastructure damage, emphasizing the importance of resilient urban planning and adaptive infrastructure systems.



04

### AID DISTRIBUTION IMBALANCE

International aid is disproportionately concentrated on high economic-loss events, with limited consideration for the scale of human impact or the number of affected individuals.



05

### EQUITY IN GLOBAL RESPONSE

Aid allocation is not fully balanced, as support is often directed to a small number of countries.

This highlights the need for more equitable and multi-factor allocation frameworks.



06

### DATA & TRANSPARENCY

Improved disaster reporting and documentation are essential to ensure faster, more accurate, and more effective international response.



07

### STRATEGIC CONCLUSION

Sustainable disaster resilience requires a combined approach:

-  Resilient Infrastructure
-  Equitable Aid Distribution
-  Strong Preparedness Systems
-  Effective Governance



### OVERALL MESSAGE

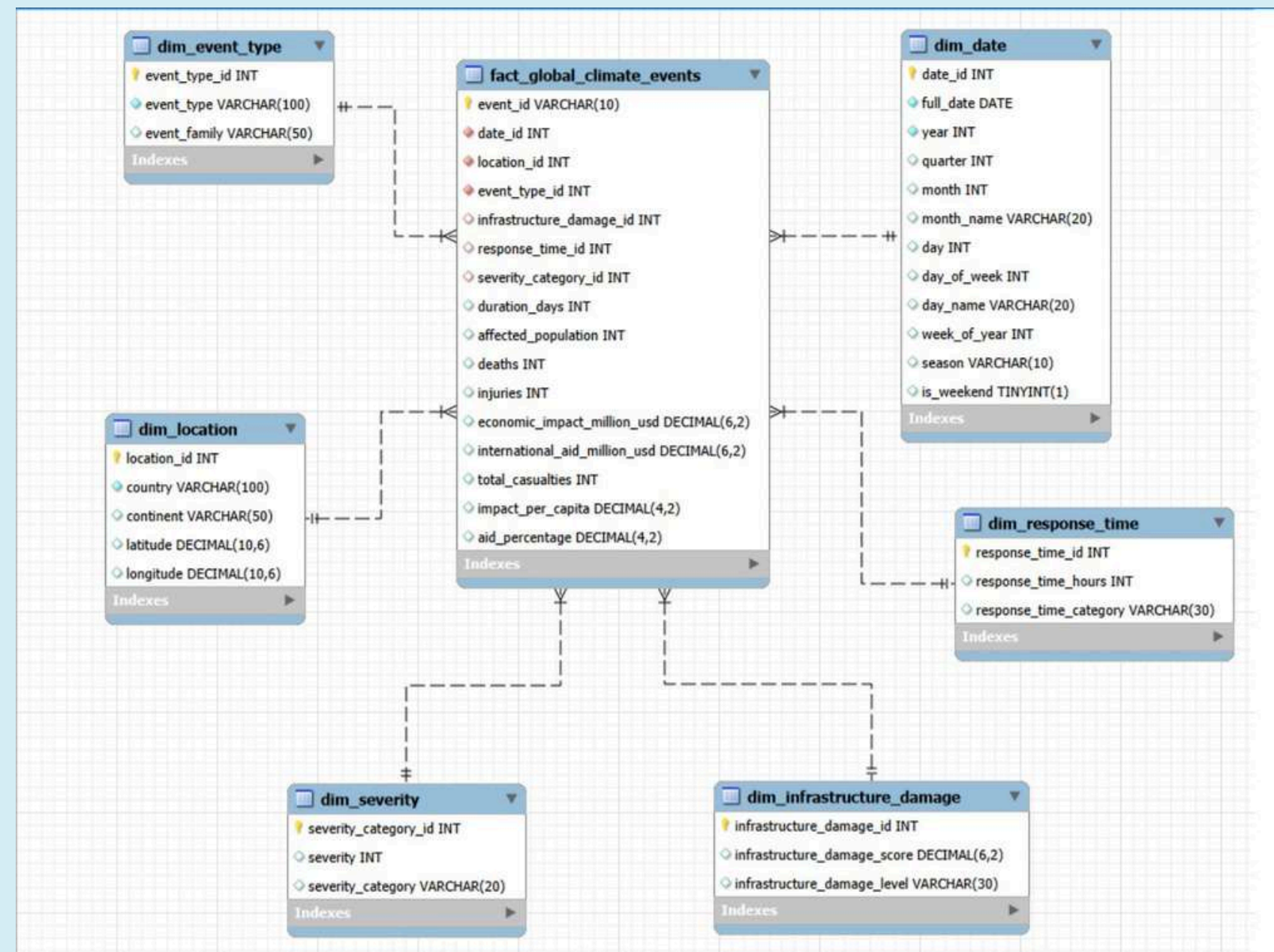
Achieving sustainable development requires not only economic growth, but also resilient infrastructure, effective disaster preparedness, equitable international support, and stronger protection for vulnerable populations.





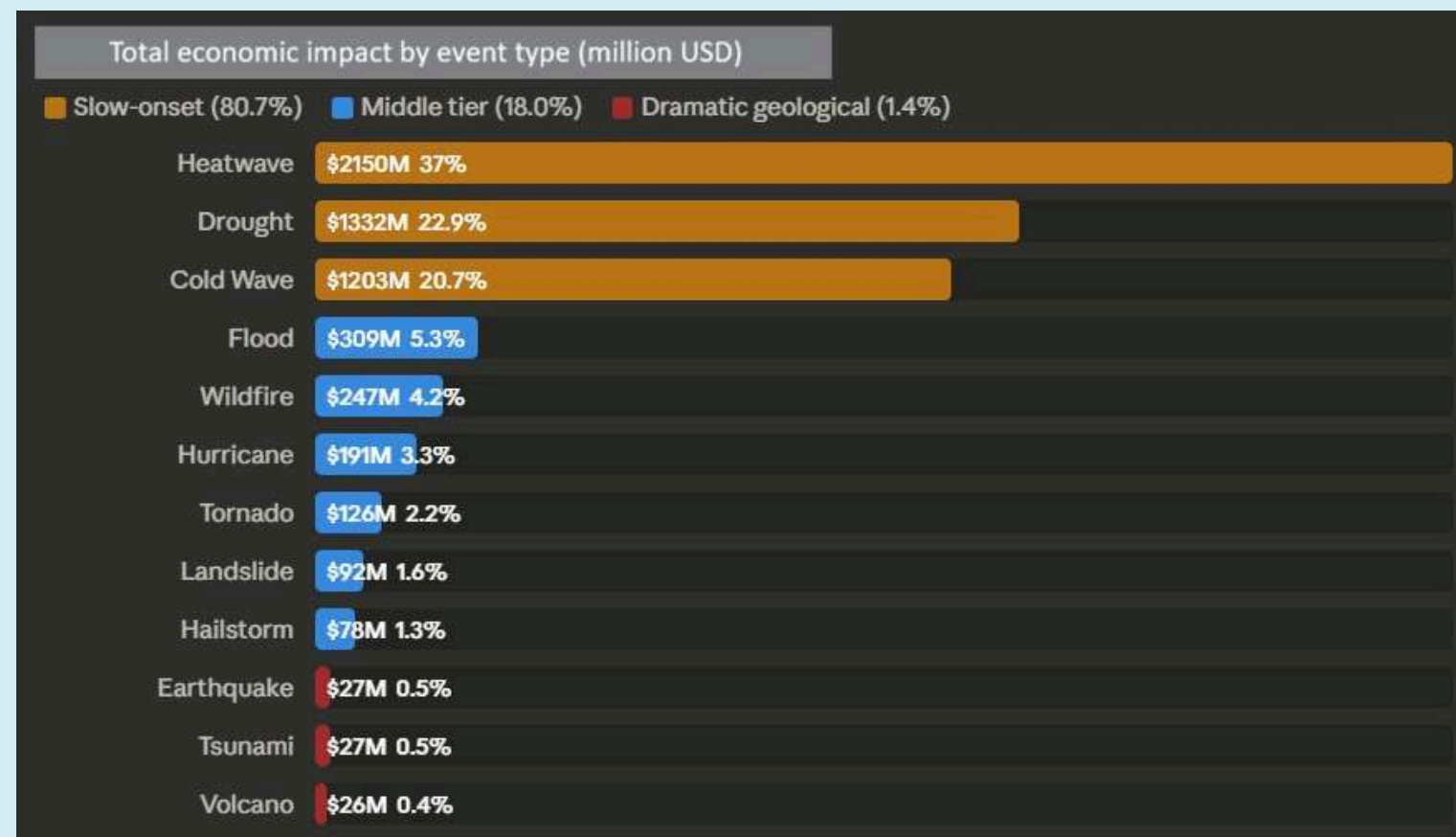
# 4- Methodology

## Data Modeling using My SQL



# 4- Methodology

## Charts & Insights



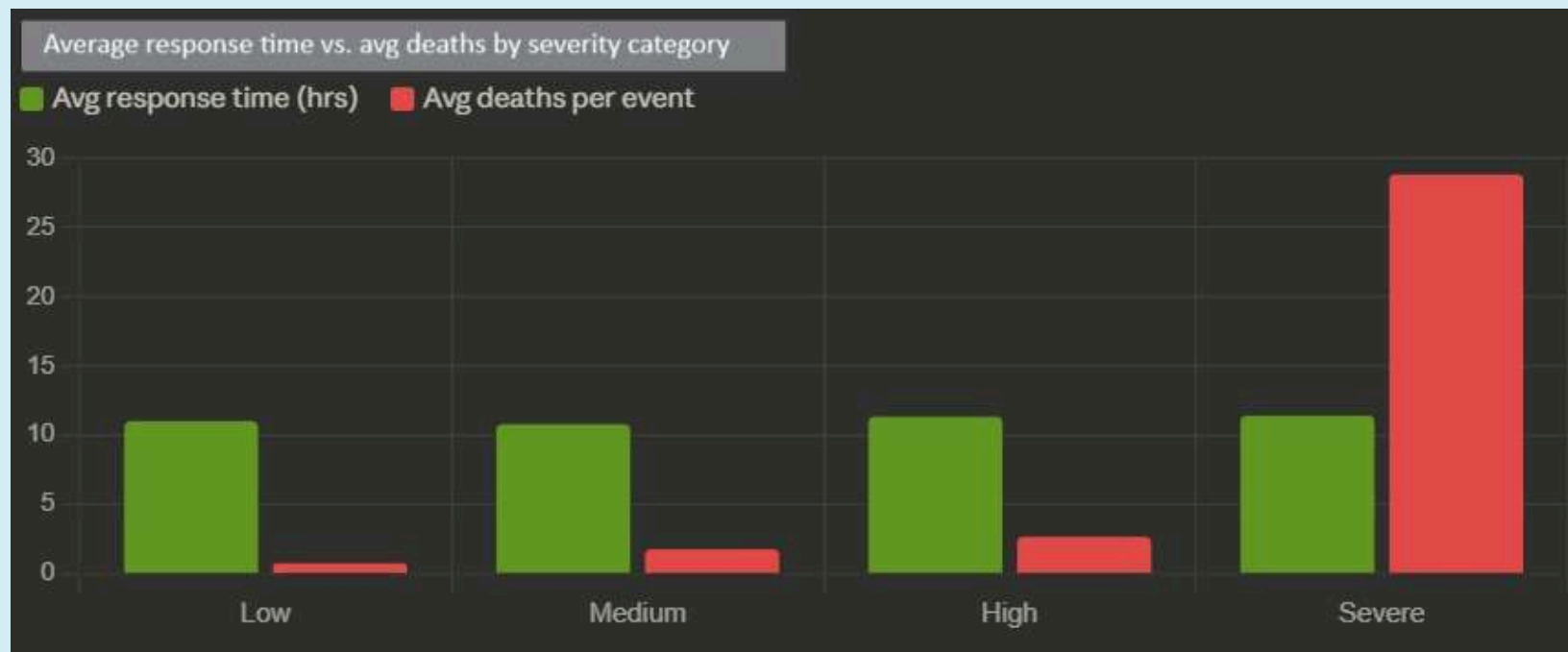
Slow-onset events dominate economic damage, Heatwaves, Droughts, and Cold Waves are not dramatic to watch no collapsing buildings, no tsunami waves yet they account for the overwhelming majority of financial destruction globally between 2020 and 2025.

|   | event_type        | event_family        | occurrence_count | total_impact_million_USD | total_deaths | avg_severity |
|---|-------------------|---------------------|------------------|--------------------------|--------------|--------------|
| ▶ | Volcanic Eruption | Geological          | 255              | 26.08                    | 397          | 2.41         |
|   | Tsunami           | Water Related       | 237              | 26.86                    | 331          | 2.46         |
|   | Earthquake        | Geological          | 282              | 27.43                    | 443          | 2.45         |
|   | Hailstorm         | Temperature Extreme | 242              | 77.93                    | 1354         | 4.43         |
|   | Landslide         | Geological          | 266              | 92.25                    | 1621         | 4.53         |
|   | Tornado           | Wind Storm          | 234              | 126.01                   | 1366         | 4.44         |
|   | Hurricane         | Wind Storm          | 245              | 190.74                   | 364          | 2.57         |
|   | Wildfire          | Fire                | 235              | 246.59                   | 972          | 4.48         |
|   | Flood             | Water Related       | 217              | 309.19                   | 1676         | 4.41         |
|   | Cold Wave         | Temperature Extreme | 233              | 1203.18                  | 1405         | 4.51         |
|   | Drought           | Temperature Extreme | 254              | 1332.43                  | 1694         | 4.52         |
|   | Heatwave          | Temperature Extreme | 229              | 2149.61                  | 1860         | 4.62         |



# 4- Methodology

## Charts & Insights

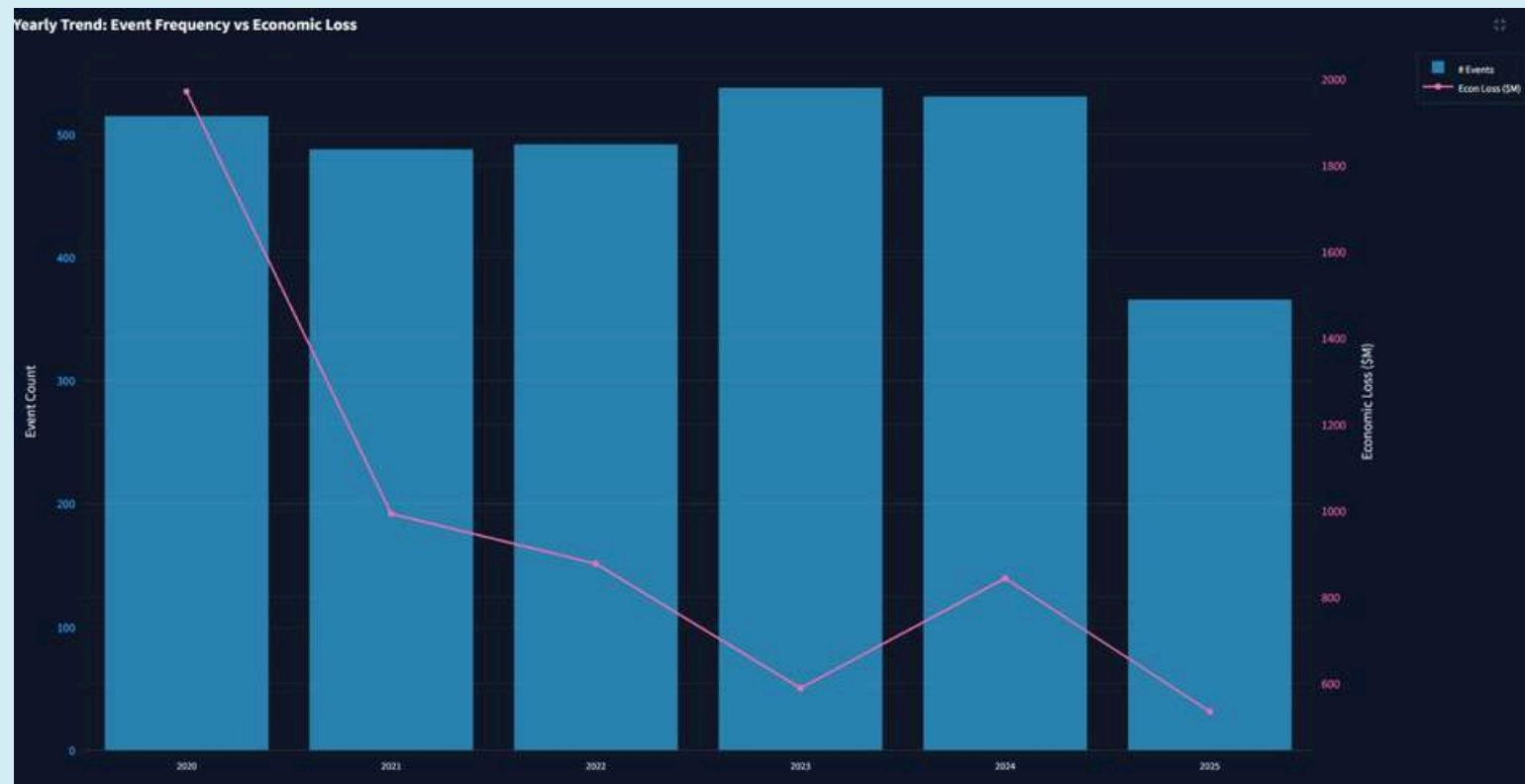
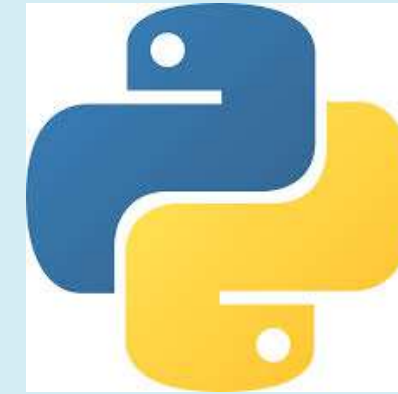


Response speed does not track severity  
Across all four severity categories, average response time stays locked in a narrow 10.8–11.4 hour band. Meanwhile, the human cost scales dramatically with severity exposing a systemic failure to prioritize mobilization for the most dangerous events.

|   | severity_category | event_count | avg_response_time_hr | min_response_hr | max_response_hr | avg_deaths | avg_economic_impact_million_USD |
|---|-------------------|-------------|----------------------|-----------------|-----------------|------------|---------------------------------|
| ► | Medium            | 962         | 10.80                | 0               | 54              | 1.77       | 1.43                            |
|   | Low               | 900         | 11.03                | 0               | 52              | 0.76       | 0.13                            |
|   | High              | 752         | 11.34                | 0               | 50              | 2.67       | 2.94                            |
|   | Severe            | 315         | 11.42                | 0               | 60              | 28.84      | 6.67                            |

# 4- Methodology

## Insights



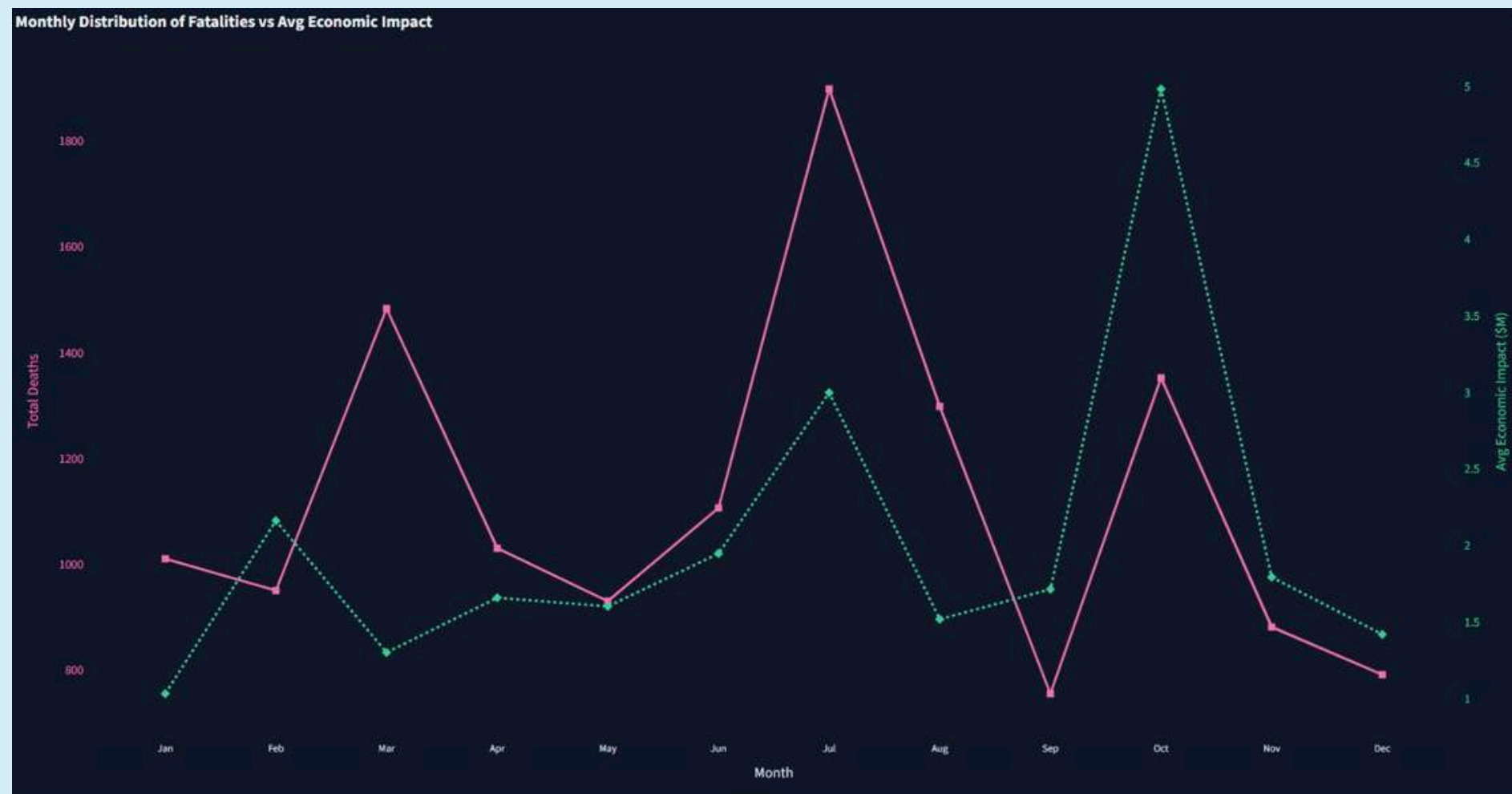
Climate events remained consistently high across the years, while economic impact showed a noticeable decline after 2020, which may indicate improved preparedness and disaster management efficiency.

The number of events in 2025 is lower because the data is only up to September 2025.



# 4- Methodology

## Insights



Climate events show seasonal patterns, with mid-year months experiencing more severe and costly disasters.

Disasters in October are relatively fewer in number but more economically destructive.

Recommendation:

Governments should strengthen disaster preparedness during high-risk months, especially July and October.



# Thanks

